

Exact height-one probabilities of three-dimensional Abelian sandpiles, spanning-tree entropies of the snub-square and Cairo lattices, and elementary resistor-network reductions

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We compute exact closed forms for three related lattice quantities: the probability P_1 that a site of the Abelian sandpile has minimal height, the spanning-tree entropy, and two-point resistances. All three are controlled by the same Kirchhoff transfer currents. In three dimensions we obtain P_1 for the simple-cubic, body-centred cubic, face-centred cubic and pyrochlore lattices as polynomials in the corresponding Watson integral W and in $1/(\pi^2 W)$. For example, on the simple cubic lattice $P_1 = \frac{4}{3}\alpha^3(2 - 3\alpha)^2 = 0.054582470221\dots$ with $\alpha = (7W + 54/(\pi^2 W))/36$. For bipartite lattices of girth at least 6 with symmetric second neighbours we prove $P_1 = (z-2)^{z-1}/[z(z-1)^{z-1}]$, which gives $1/12$ for the honeycomb (known) and the Laves graph, and $2/27$ for diamond. In two dimensions we give exact P_1 for the king, 4.8.8, Union Jack, star, dice, Lieb, snub-square, checkerboard, triakis and next-nearest-neighbour honeycomb lattices. The spanning-tree entropy of the snub-square (Shastry–Sutherland) lattice is $z = \frac{10}{3\pi}G + \frac{1}{3}\ln(2 + \sqrt{3})$, and that of its dual, the Cairo pentagonal lattice, is $\frac{20}{9\pi}G + \frac{2}{9}\ln(2 + \sqrt{3})$, where G is Catalan’s constant. The Shastry–Sutherland network with arbitrary dimer conductance has elementary resistances and a Clausen-function entropy. The king lattice has $z = 20G/(3\pi)$. We give a single elementary formula for the axis-bond resistance of any square lattice with diagonal bonds, and we reduce next-nearest-neighbour networks on bipartite girth- ≥ 6 lattices to one Green function (elliptic for the honeycomb, Joyce’s FCC form for diamond). Every result was identified by PSLQ at 25–60 digits and confirmed independently by exact finite-volume computations or Monte Carlo simulation. An appendix gives a Hurwitz-zeta closed form for the added mass of a sphere touching the inside of a spherical container.

I. INTRODUCTION

The Abelian sandpile model [1, 2] is the standard exactly solvable model of self-organized criticality. Majumdar and Dhar showed that the probability P_1 that a site has the minimal height in the stationary state is a finite determinant of lattice Green functions [3]. On the square lattice this gives $P_1 = 2/\pi^2 - 4/\pi^3$. Priezzhev computed the higher heights [4]. Exact values have since been obtained on the honeycomb [5], triangular and hexagonal [6], and, very recently, kagome [7] lattices. The Majumdar–Dhar method applies in any dimension. Nevertheless, to our knowledge no closed form for P_1 on a three-dimensional lattice has appeared. The obstacle is the need for exact resistances beyond nearest neighbours.

We remove that obstacle by identifying every needed resistance in the natural basis $\{1, W, 1/(\pi^2 W)\}$ of the relevant Watson integral [8, 9]. The same machinery, based on Kirchhoff’s transfer currents [10] and their spanning-tree interpretation [11, 12], also gives new spanning-tree entropies. Numerical values for many Archimedean lattices are known from Chang and Wang [13], Chang and Shrock [14], and Guttmann and Rogers [15]; we add closed forms. It also gives exact resistances on networks whose resistances were previously known only numerically [16] or via mappings [17].

Status of novelty. All statements were checked by web search against the literature. The honeycomb and kagome values of P_1 are known [5, 7]; we reproduce them as validation. The king-lattice entropy $20G/(3\pi)$ may be

implicit in the Mahler-measure literature [15]. For the remaining results our searches found no prior closed form. Novelty should be read in that limited sense.

Notation

All resistors are 1Ω unless a conductance is specified, and $R(x)$ is the effective resistance between the origin and x . For an arc $a = (o \rightarrow h_a)$ the transfer current is $Y_{ab} = \frac{1}{2}[R(h_a) + R(h_b) - R(h_a - h_b)]$. G denotes Catalan’s constant, and $\text{Cl}_2(\theta) = \sum_{n \geq 1} \sin(n\theta)/n^2$ is the Clausen function. Complete elliptic integrals use the parameter convention $K(m) = \int_0^{\pi/2} (1 - m \sin^2 \theta)^{-1/2} d\theta$. $\mathbb{E}_k$ denotes the normalized Brillouin-zone average.

II. ABELIAN SANDPILES ON 3D LATTICES

A. Method

In the Abelian sandpile on a graph, a site v topples when its height exceeds $\deg(v)$ and sends one grain to each neighbour. Majumdar and Dhar [3] showed that

$$P_1 = \det(I - M), \quad M_{jk} = D(x_j) + D(x_k) - D(x_j - x_k), \quad (1)$$

where $D = R/2$ and x_j runs over the $z - 1$ neighbours of the origin whose bonds are removed by the defect (one

bond is kept). The matrix M is exactly the transfer-current matrix Y restricted to $z - 1$ arcs. Hence

$$P_1 = \det(I - Y'), \quad (2)$$

with Y' equal to Y minus one arc. An internal consistency check is that P_1 must not depend on which bond is kept. The check is nontrivial when the bonds at a site are inequivalent, and it holds for every result below.

Corollary (UST leaves). Since P_1 is independent of the kept bond, the probability that a degree- z vertex is a leaf of the uniform spanning tree is exactly zP_1 . On the square lattice this reproduces the known value $(8/\pi^2)(1 - 2/\pi)$.

B. Watson integrals

Table I lists the three constants we need. For FCC we use $W_f = \mathbb{E}[1/(3 - s)]$, which is one third of Watson's normalization $\mathbb{E}[1/(1 - s/3)]$.

C. Resistances and results

In each case below, $q \equiv 1/(\pi^2 W)$ with W the lattice's own Watson integral.

Simple cubic ($z = 6$). The resistances are $R(1, 0, 0) = 1/3$, $R(1, 1, 0) = \alpha$ and $R(2, 0, 0) = 2 - 4\alpha$, with

$$\alpha = \frac{7W + 54/(\pi^2 W)}{36} = 0.395079152341851372 \dots \quad (3)$$

The 5×5 determinant factorizes:

$$P_1^{\text{SC}} = \frac{4}{3} \alpha^3 (2 - 3\alpha)^2 = 0.054582470221053656104 \dots \quad (4)$$

Body-centred cubic ($z = 8$, neighbours $(\pm 1, \pm 1, \pm 1)$). The resistances are $R(1, 1, 1) = 1/4$, $R(2, 0, 0) = (W - 4q)/4$, $R(2, 2, 0) = 4q$ and $R(2, 2, 2) = (8 - 3W - 36q)/4$. Then

$$\begin{aligned} P_1^{\text{BCC}} &= \frac{3q}{128} (W + 4q)^3 (4 - W - 12q)^3 \\ &= 0.042454248763323867910 \dots \end{aligned} \quad (5)$$

Face-centred cubic ($z = 12$, neighbours are the permutations of $(\pm 1, \pm 1, 0)$). The resistances are $R(1, 1, 0) = 1/6$, $R(2, 0, 0) = (4W - 3q)/6$, $R(2, 1, 1) = (2W + 3q - 1)/3$ and $R(2, 2, 0) = (8 - 12W - 9q)/3$. Then

$$\begin{aligned} P_1^{\text{FCC}} &= 6^{-10} (5 - 4W - 6q)^2 (7 - 8W - 3q)^3 \\ &\quad \times (8W + 3q + 1)^3 (16W + 15q - 5)^3 \\ &= 0.029126520511690534744 \dots \end{aligned} \quad (6)$$

Pyrochlore ($z = 6$, corner-sharing tetrahedra, $W = W_f$, $X = 1/(\pi^2 W_f)$). The Laplacian transfer currents from an edge are $1/3$ (self), $1/6$ (the two other edges of the same tetrahedron), $(1 + 8W + 3X)/48$ (collinear

partner) and $(15 - 8W - 3X)/96$ (the other two). Then Eq. (2) gives a perfect square:

$$\begin{aligned} P_1^{\text{pyro}} &= \left[\frac{(31 - 8W - 3X)(67 + 24W + 9X)}{9216} \right]^2 \\ &= 0.053582950138955615678 \dots \end{aligned} \quad (7)$$

Diamond follows from the general theorem of Sec. III A: $P_1 = 2/27$.

D. Independent verification

We computed P_1 *without sampling* at the centre of finite boxes with dissipative boundaries, by exact linear algebra (no sampling). The sequences converge to the formulas:

Lattice	Boxes	Exact finite-box P_1
SC	$N = 16, 24, 32$	0.0546206, 0.0545941, 0.0545875
BCC	$N = 16, 24, 32$	0.0424902, 0.0424653, 0.0424590
FCC	$N = 12, 18, 24$	0.0291795, 0.0291427, 0.0291335
FCC	$N = 30$	0.0291301
Pyrochl.	$N = 10, 16, 22$	0.0536309, 0.0535949, 0.0535876
Diamond	N increasing	0.0742235, 0.0741125, 0.0740859
Laves	$N = 10, 16, 24$	0.0834488, 0.0833615, 0.0833418

Monte Carlo simulations on 32^3 – 44^3 lattices give SC 0.05451(23), BCC 0.04244(21) and FCC 0.02926(8). An earlier FCC run gave 0.02865(17), which we attribute to slow mixing; the exact finite-box sequence resolves the discrepancy in favour of the formula. The corresponding UST leaf probabilities zP_1 are SC 0.3274948, BCC 0.3396340, FCC 0.3495182, pyrochlore 0.3214977 and diamond 8/27.

III. EXACT $P(h = 1)$ CALCULATIONS

A. A general theorem for bipartite lattices

Theorem 1. Let the lattice be z -regular, bipartite, of girth at least 6 (so that $A^2 = zI + A_2$), and edge-transitive, and let its second-neighbour graph A_2 also be edge-transitive. Then $R(\text{nearest}) = 2/z$, $R(\text{second}) = 2/(z - 1)$, and

$$P_1 = \frac{(z - 2)^{z-1}}{z(z - 1)^{z-1}}. \quad (8)$$

Proof. Write $L_2 = z^2 I - A^2 = LQ$. This operator acts within each sublattice, so $R_L(u, v) = z R_{L_2}(u, v)$ for u, v on the same sublattice. Foster's theorem [18] on the edge-transitive second-neighbour graph gives $R_{L_2} = 2/[z(z - 1)]$ across a second-neighbour bond. Hence Y' has the constant diagonal $2/z$ and constant off-diagonal entries. Its determinant then factorizes into Eq. (8). $\square$

The theorem gives $1/12$ for the honeycomb, which is known [5], and $1/12$ for the Laves graph (srs). It

TABLE I. Watson integrals ($c_i = \cos k_i$, $s = c_1 c_2 + c_2 c_3 + c_3 c_1$).

Lattice	Definition and closed form	Value
SC	$\mathbb{E}\left[\frac{1}{1-(c_1+c_2+c_3)/3}\right] = \frac{\sqrt{6}\Gamma(\frac{1}{24})\Gamma(\frac{5}{24})\Gamma(\frac{7}{24})\Gamma(\frac{11}{24})}{32\pi^3}$	1.51638605915
BCC	$\mathbb{E}\left[\frac{1}{1-c_1 c_2 c_3}\right] = \frac{\Gamma(\frac{1}{4})^4}{4\pi^3}$	1.39320392969
FCC	$\mathbb{E}\left[\frac{1}{3-s}\right] = \frac{3\Gamma(\frac{1}{3})^6}{2^{14/3}\pi^4}$	0.44822039439

gives $2/27 = 0.0740740\dots$ for diamond and applies to d -dimensional diamond with $z = d + 1$.

B. The king lattice

Take $\mathbb{Z}^2$ with all eight king-move neighbours ($z = 8$). Put $p = 1/\pi$ and $L = \ln(2 + \sqrt{3})/(\pi\sqrt{3}) = 0.24202564754285515484$. The resistances are

$$\begin{aligned} R(1,0) &= L, & R(1,1) &= \frac{1}{2} - L, \\ R(2,0) &= \frac{4}{\pi} - 4L, & R(2,1) &= 6L - \frac{1}{2} - \frac{2}{\pi}, \\ R(2,2) &= 5 - \frac{4}{\pi} - 14L, \end{aligned} \quad (9)$$

and the 7×7 determinant factorizes as

$$\begin{aligned} P_1^{\text{king}} &= -\frac{1}{2048}(3 + 4p - 14L)(1 + 4p - 2L)(6L + 12p - 7) \\ &\quad \times (13 - 32p + 48p^2 - 52L - 12L^2)^2 \\ &= 0.0420560364703691189817\dots \end{aligned} \quad (10)$$

A Monte Carlo simulation on 96×96 gives 0.04218 ± 0.00033 .

C. Further two-dimensional lattices

Table II collects the remaining exact values. We write $c = \sqrt{2} \arccos(1/3)/\pi = 0.55412642397957198712$ for the $4.8.8/\text{Union Jack}$ constant, $p = 1/\pi$ and $r = \sqrt{3}/\pi$.

Honeycomb with next-nearest neighbours (unit weights, $z = 9$). Let $g = 4K(64/189)/(3\sqrt{21}\pi) = 0.16088977460548913890$, the triangular Green function at $t = 13/2$, and let $g' = dg/dt$. All resistances among the site and its nine neighbours are rational combinations of $1, r, g, g'$, and

$$\begin{aligned} P_1 &= \frac{(2 - 7g)(12r - 14 - 105g - 1120g') \mathcal{P}^2}{196036848}, \\ \mathcal{P} &= 729120gg' + 1764gr^2 - 7224gr + 9163g + 8202600g'^2 \\ &\quad - 136080g'r + 305760g' + 792r^2 - 2688r + 2548, \end{aligned} \quad (11)$$

which evaluates to $P_1 = 0.037899503465150422174$. Exact finite boxes of 40^2 , 80^2 and 160^2 cells give 0.0379335 , 0.0379082 and 0.0379017 .

Validation against the literature. The same pipeline gives the kagome value $(5 + 3r)(17 - 6r)/1296$, which

matches the value reported in Ref. [7]. It gives $1/12$ for the honeycomb [5] and reproduces the square lattice's $2/\pi^2 - 4/\pi^3$ to 20 digits. Monte Carlo checks on open patches: $4.8.8$ $0.0807\text{--}0.0817 (\pm 0.0006)$; star $0.0697\text{--}0.0706 (\pm 0.0008)$; dice hub $0.01273(30)$, rim $0.1888(10)$; Union Jack corner $0.02024(30)$, centre $0.1535(8)$. The checkerboard and triakis values were confirmed by exact finite boxes ($40^2, 80^2, 160^2$ cells): 0.0535660 , 0.0535281 , 0.0535185 and 0.0035988 , 0.0035940 , 0.0035928 .

IV. SPANNING-TREE ENTROPY ON CAIRO AND SNUB-SQUARE LATTICES

The number of spanning trees of an N -vertex piece of a lattice grows as e^{zN} [19]. For a periodic lattice, z is $|V_{\text{cell}}|^{-1} \mathbb{E}_k \ln \det L(k)$.

A. Snub square and Cairo

For the snub square tiling $3^2.4.3.4$ (the Shastri–Sutherland lattice, 4 vertices per cell),

$$\begin{aligned} \det L(k) &= 2(170 - 72(\cos k_1 + \cos k_2) - 28 \cos k_1 \cos k_2 \\ &\quad + \cos 2k_1 + \cos 2k_2). \end{aligned} \quad (12)$$

We find

$$\mathbb{E}_k \ln \det L = \frac{40}{3\pi}G + \frac{4}{3} \ln(2 + \sqrt{3}), \quad (13)$$

and hence

$$\begin{aligned} z_{\text{snub}} &= \frac{10}{3\pi}G + \frac{1}{3} \ln(2 + \sqrt{3}) \\ &= 1.41085564574433483667\dots, \\ z_{\text{Cairo}} &= \frac{20}{9\pi}G + \frac{2}{9} \ln(2 + \sqrt{3}) \\ &= 0.94057043049622322445\dots \end{aligned} \quad (14)$$

The Cairo pentagonal tiling is the planar dual. It has the same number of trees per cell but 6 vertices per cell.

Why it closes. The z_2 -discriminant of the spectral curve $\det L = 0$ has only one simple factor, $z_1^2 - 98z_1 + 1$, whose roots are $(\sqrt{3} \pm \sqrt{2})^4$. After the node at $k = 0$ is removed, the curve therefore has genus zero. The entropy must then be a combination of dilogarithms at algebraic points.

TABLE II. Exact height-one probabilities P_1 of the Abelian sandpile. Each value is independent of which bond is kept in Eq. (1).

Lattice (site, degree)	P_1	Value
4.8.8 (3)	$c(4 - 3c)/16$	0.08095858841632017693
Union Jack, face centre (4)	$9c^2(1 - c)/8$	0.15402172338027135905
Union Jack, corner (8)	$\frac{1}{256}(9c - 2 - 4p)(6 - 7c - 4p)(7 - 9c - 4p)(81c^2 - 105c + 34 - 4p - 16p^2)^2$	0.02043343799324922073
Star 3.12.12 (3)	$7/100$	0.07
Dice, rim (3)	$3/16$	0.1875
Dice, hub (6)	$[(1 + 2r)(11 - 6r)/144]^2$	0.01261518916334840149
Lieb, corner (4)	$(\pi - 2)/(4\pi^3)$	0.00920452869398469827
Lieb, edge (2)	$1/4$	0.25
Snub square (5)	$\frac{8953}{3888} - \frac{281\sqrt{3}}{216} + (\frac{44\sqrt{3}}{27} - \frac{11}{4})p + (\frac{17}{36} - \frac{\sqrt{3}}{3})p^2$	0.06191455535342906165
Checkerboard (6)	$13(7 - 8p)(11 + 8p)(1 + 16p - 16p^2)/65536$	0.05351525370368361765
Triakis, lattice vertex (12)	$(6r - 1)(5 + 18r)^2(41 - 42r)^2/45562500$	0.00359238258113540057
Triakis, triangle centre (3)	$27/100$	0.27
Kagome (4) [known [7]]	$(5 + 3r)(17 - 6r)/1296$	0.07029827383269397637
Laves graph (3)	$1/12$	0.083
Diamond (4)	$2/27$	0.074

PSLQ over $\{\ln 2, \ln 3, \ln(1 + \sqrt{2}), \ln(2 + \sqrt{3}), \ln(\sqrt{2} + \sqrt{3}), G/\pi, \text{Cl}_2(\pi/3)/\pi, \text{Cl}_2(\pi/12)/\pi, \text{Cl}_2(5\pi/12)/\pi\}$ returned $-3\mathbb{E} \ln \det L + 4\ln(2 + \sqrt{3}) + 40G/\pi = 0$ [25] with residual below 10^{-58} at 60 digits. The numerical value 1.41086 is known [13]; we found no prior closed form.

The same genus-zero structure gives *algebraic* resistances (Table III): $R_{\text{tri-sq}} = \frac{1}{2} - \sqrt{3}/18$ and $R_{\text{tri-tri}} = 2\sqrt{3}/9$. Foster's sum rule $8(\frac{1}{2} - \sqrt{3}/18) + 2(2\sqrt{3}/9) = 4$ holds exactly. Planar duality, $R(e) + R(e^*) = 1$ (an edge is in the spanning tree exactly when its dual edge is not in the dual tree), then gives the Cairo values $\frac{1}{2} + \sqrt{3}/18$ and $1 - 2\sqrt{3}/9$. The remaining transfer currents at a snub-square vertex are $Y(a, b) = 1 - \sqrt{3}/2$, $Y(a, d) = \sqrt{3}/9$, $Y(a', b') = (3 - \sqrt{3})/6$, $Y(d, a') = \frac{1}{2} - 2\sqrt{3}/9$, $Y(a, a') = 5\sqrt{3}/9 - 1 + 1/(2\pi)$ and $Y(a, b') = \frac{1}{2} - \sqrt{3}/9 - 1/(2\pi)$. Here d is the dimer arc; a, b each form a triangle with d ; a', b' form a triangle with each other; a' is collinear with a and b' with b .

Then

$$R_{\text{dimer}} = \frac{2\Phi}{\pi\sqrt{w(w+2)}}, \quad (16)$$

$$R_{\text{square}} = \frac{1}{2} - \frac{\sqrt{w}\Phi}{2\pi\sqrt{w+2}}, \quad (17)$$

$$z(w) = \frac{1}{\pi} \left[2G + \Phi \operatorname{arccosh}(w+1) + \text{Cl}_2\left(\frac{\pi}{2} + \Phi\right) + \text{Cl}_2\left(\frac{\pi}{2} - \Phi\right) \right]. \quad (18)$$

Derivation. Start from $z(0) = 4G/\pi$. Kirchhoff's relation $\partial z/\partial w = \frac{1}{2}R_{\text{dimer}}(w)$ (the derivative of the log tree count with respect to a conductance is the sum of the corresponding edge resistances) then integrates via $u + 1 = \sec \phi$ and

$$\int_0^\Phi \phi \sec \phi d\phi = \Phi \ln \tan\left(\frac{\pi}{4} + \frac{\Phi}{2}\right) - 2G + \text{Cl}_2\left(\frac{\pi}{2} + \Phi\right) + \text{Cl}_2\left(\frac{\pi}{2} - \Phi\right). \quad (19)$$

B. Shastry–Sutherland network with arbitrary dimer conductance

Give the square bonds conductance 1 and the dimer bonds conductance $w \geq 0$, and let $\Phi = \arccos(1/(w+1))$.

Special cases. At $w = 1$, $\Phi = \pi/3$, and $\text{Cl}_2(\pi/6) + \text{Cl}_2(5\pi/6) = 4G/3$ recovers the snub-square values. At $w = 2$, $z(2) = 1.57336855075766435824$, which coincides with the independently computed Union Jack entropy per vertex. As $w \rightarrow \infty$, $z = \frac{1}{2} \ln(2w) + 2G/\pi + o(1)$.

C. The king lattice

Take $\mathbb{Z}^2$ with unit axis bonds and diagonal conductance w , and let $\varphi = \arccos(2w)$. Then

$$R_{\text{axis}} = \frac{\arccos(2w)}{\pi\sqrt{1-4w^2}}, \quad (20)$$

$$R_{\text{diag}} = \frac{1}{2w} - \frac{\arccos(2w)}{\pi w\sqrt{1-4w^2}}, \quad (21)$$

for $w < \frac{1}{2}$. For $w > \frac{1}{2}$, use the continuation $\operatorname{arccosh}(2w)/\sqrt{4w^2-1}$. The entropy is

$$z(w) = \frac{4G}{\pi} + \ln w + \frac{2}{\pi} \left[\varphi \ln \tan\left(\frac{\pi}{4} + \frac{\varphi}{2}\right) + \operatorname{Cl}_2\left(\frac{\pi}{2} + \varphi\right) + \operatorname{Cl}_2\left(\frac{\pi}{2} - \varphi\right) \right], \quad (22)$$

and at $w = 1$ (the plain king graph)

$$z_{\text{king}} = \frac{20G}{3\pi} = 1.94373936020545853426\dots, \quad (23)$$

with $R_{\text{axis}} = \ln(2 + \sqrt{3})/(\pi\sqrt{3})$ and $R_{\text{diag}} = \frac{1}{2} - R_{\text{axis}}$. The determinant $9 - (1 + x + x^{-1})(1 + y + y^{-1})$ belongs to a family of Mahler measures, so $20G/(3\pi)$ may be implicit in that literature [15]. The elementary structure has a simple origin: after the exact k_2 average and the substitution $x = -1 + t^2$, the remaining integrand has discriminant $8w + (2w - 1)^2 = (2w + 1)^2$, a perfect square.

Corollaries. (i) Eliminating each face centre of the Union Jack lattice with centre-bond conductance w (star-mesh) yields the king lattice with axis conductance $1 + w/2$ and diagonal ratio $w' = w/(4 + 2w)$. At $w = 1$ this reproduces $R_{\text{corner-corner}} = c/2$. (ii) The checkerboard lattice with crossed-diagonal conductance w has $\det = 4(w + 1)[2(a + b) + (w - 1)ab]$, with $a = 1 - \cos(k_1 + k_2)$ and $b = 1 - \cos(k_1 - k_2)$. This is the king determinant with $w' = (1 - w)/(2(1 + w))$.

V. RESISTOR NETWORK REDUCTIONS

A. Adjacent resistances on Archimedean and Laves tilings

The 4.8.8 and Union Jack lattices are dual, and the entries in Table III satisfy $R(e) + R(e^*) = 1$ identically. The 4.8.8 determinant factorizes as $64 - 4(3 + \cos k_1)(3 + \cos k_2)$, which is the origin of $\arccos(1/3)$. Owaidat computed Union Jack resistances numerically [16]; Cserti *et al.* gave exact mappings for kagome, dice and decorated lattices [17]. Kirchhoff solves on 120×120 -cell tori, Richardson-extrapolated, agree with the table to about 10^{-7} .

B. Square lattice with arbitrary diagonal bonds

Give conductance w_1 to bonds along e_1 , w_2 along e_2 , w_+ along $e_1 + e_2$ and w_- along $e_1 - e_2$. Let $\sigma = w_+ + w_-$

TABLE III. Exact adjacent-node resistances (1Ω resistors). $c = \sqrt{2} \arccos(1/3)/\pi$. Pairs related by planar duality satisfy $R(e) + R(e^*) = 1$.

Tiling, edge	R	Value
Snub square, tri-sq	$\frac{1}{2} - \frac{\sqrt{3}}{18}$	0.40377495513506237
Snub square, tri-tri	$\frac{2\sqrt{3}}{9}$	0.38490017945975051
Cairo, dual of tri-sq	$\frac{1}{2} + \frac{\sqrt{3}}{18}$	0.59622504486493763
Cairo, dual of tri-tri	$1 - \frac{2\sqrt{3}}{9}$	0.61509982054024949
4.8.8, square	$\frac{1}{2} + \frac{c}{4}$	0.63853160599489299
4.8.8, oct-oct	$1 - \frac{c}{2}$	0.72293678801021400
Union Jack, corner-corner	$\frac{c}{2}$	0.27706321198978599
Union Jack, corner-centre	$\frac{2-c}{4}$	0.36146839400510700
King, axis	$\frac{\ln(2+\sqrt{3})}{\pi\sqrt{3}}$	0.24202564754285515
King, diagonal	$\frac{1}{2} - \frac{\ln(2+\sqrt{3})}{\pi\sqrt{3}}$	0.25797435245714485

and $\Delta = 4w_+w_- - w_1^2$. Then

$$R(e_1) = \begin{cases} \frac{2}{\pi\sqrt{\Delta}} \operatorname{arcsinh} \sqrt{\frac{\Delta}{(w_1 + \sigma)(w_1 + w_2)}}, & \Delta > 0, \\ \frac{2}{\pi\sqrt{-\Delta}} \operatorname{arcsin} \sqrt{\frac{-\Delta}{(w_1 + \sigma)(w_1 + w_2)}}, & \Delta < 0, \\ \frac{2}{\pi\sqrt{(w_1 + \sigma)(w_1 + w_2)}}, & \Delta = 0. \end{cases} \quad (24)$$

Derivation. At fixed $x = \cos k_1$ the k_2 average of $1/\det$ is $1/\sqrt{A^2 - M^2}$ with $A^2 - M^2 = 4(1 - x)(p + qx)$, where $p = (w_1 + w_2 + \sigma)^2 - w_2^2 - (w_+ - w_-)^2$ and $q = \Delta$. The factor $1 - x$ is the node at $k = 0$. Then $R(e_1) = \pi^{-1} \int_{-1}^1 dx / \sqrt{(1 + x)(p + qx)}$, and $p - q = 2(w_1 + \sigma)(w_1 + w_2)$. *Special cases.* The square lattice gives $1/2$; the king lattice gives $\arccos(2w)$ as above; and the triangular lattice ($w_- = 0$) gives Kenyon's isoradial value [20], $1/3$ in the isotropic case. Together with Foster's identity $w_1 R(e_1) + w_2 R(e_2) + w_+ R(e_1 + e_2) + w_- R(e_1 - e_2) = 1$, this also gives the diagonal resistances when $w_+ = w_-$. The formula agrees with direct 2D integration to $\sim 10^{-26}$ for random conductances.

C. Next-nearest-neighbour networks on bipartite lattices

Let the lattice be z -regular, bipartite and of girth at least 6, with unit nearest-neighbour bonds and conductance w on second neighbours. Since $A^2 = zI + A_2$,

$$L = (z + z(z - 1)w)I - A - w(A^2 - zI), \quad (25)$$

$$\det = u(1 + 2zw + w^2u), \quad u = z^2 - |f(k)|^2. \quad (26)$$

The symmetry identity $\mathbb{E}[\text{Ref}/\det] = \mathbb{E}[|f|^2/(z\det)]$ collapses the nearest-neighbour resistance to

$$R_{\text{NN}}(w) = 2\left(w + \frac{1}{z}\right) \mathbb{E}_k \left[\frac{1}{1 + 2zw + w^2(z^2 - |f(k)|^2)} \right], \quad (27)$$

$$R_{\text{NNN}}(w) = \frac{2 - zR_{\text{NN}}}{z(z-1)w}. \quad (28)$$

Honeycomb ($z = 3$, $|f|^2 = 3 + 2S$, $S = \cos k_1 + \cos k_2 + \cos(k_1 - k_2)$). Here $\det = 2T(6w + 1 + 2w^2T)$ with $T = 3 - S$. Horiguchi's formula [21] with $e = 3 + 1/w$ gives

$$m = \frac{16w^3(3w + 1)}{(2w + 1)^3(6w + 1)}, \quad (29)$$

$$R_{\text{NN}} = \frac{4(3w + 1)K(m)}{3\pi(2w + 1)^{3/2}\sqrt{6w + 1}}, \quad (30)$$

and $R_{\text{NNN}} = (2 - 3R_{\text{NN}})/(6w)$. At $w = 1$: $m = 64/189$, $R_{\text{NN}} = 16K(64/189)/(9\sqrt{21}\pi) = 0.21451969947398551853$ and $R_{\text{NNN}} = 0.22607348359634057407$. *Diamond* ($z = 4$). Here

$$R_{\text{NN}} = \frac{4w + 1}{8w^2} P_{\text{FCC}}\left(3 + \frac{8w + 1}{4w^2}\right), \quad (31)$$

where $P_{\text{FCC}}(t) = \mathbb{E}[1/(t - s)]$ is the FCC Green function in Joyce's closed form [22]. At $w = 1$ ($t = 21/4$), $R_{\text{NN}} = 0.12347073437196593577$ and $R_{\text{NNN}} = (1 - 2R_{\text{NN}})/(6w) = 0.12550975520934468$. Three-dimensional Kirchhoff solves on 12^3 , 16^3 and 20^3 tori give $0.1234547 \rightarrow 0.1234640 \rightarrow 0.1234673$.

VI. DISCUSSION AND CONCLUSION

Three observations organize the results. First, the Majumdar–Dhar matrix *is* the transfer-current matrix, Eq. (2). Sandpile height probabilities, UST leaf and degree statistics, and resistances therefore all come from the same $z \times z$ matrix at a vertex. Second, in three dimensions every needed resistance lies in the three-dimensional space spanned by $\{1, W, 1/(\pi^2 W)\}$. This is why P_1 becomes a polynomial in the Watson integral, and why the determinants factor so strikingly. Third, in two dimensions, closed forms arise whenever the spectral curve has genus zero after removal of the node at $k = 0$. That gives algebraic or elementary resistances and Clausen-type entropies (snub square, Cairo, Shastry–Sutherland, king). Otherwise complete elliptic integrals appear (next-nearest-neighbour honeycomb).

Open questions include the higher height probabilities $P_2, P_3, \dots$ in three dimensions, the ruby and snub-hexagonal lattices (for which PSLQ found no closed form in natural bases), and a conceptual explanation of the perfect-square factorization of the pyrochlore P_1 .

Appendix A: Added mass of a sphere touching a spherical container

A companion image-sum result is recorded here because it has the same flavour: an infinite image series collapses to a single special value. Consider a ball of radius a that touches, from inside, a fixed spherical tank of radius $b > a$ filled with ideal (inviscid, incompressible, irrotational) fluid. The ball moves along the line of centres, and its added mass is $M = C\rho\frac{4}{3}\pi a^3$. With $r = a/b$ and $q = 1/(1 - r)$,

$$C(r) = \frac{3}{2} \sum_{k=0}^{\infty} \frac{1}{[1 + k(1 - r)]^3} - 1 = \frac{3}{2} q^3 \zeta(3, q) - 1, \quad (\text{A1})$$

where $\zeta(s, q)$ is the Hurwitz zeta function. More generally, let the ball touch a sphere of *signed* radius b ($b > 0$ a container, $b = \infty$ a plane, $b < 0$ an external ball of radius $|b|$) and set $\lambda = 1 - a/b$. Then $C = \frac{3}{2} \sum_{k \geq 0} (1 + k\lambda)^{-3} - 1$. Special cases: a flat wall gives $\frac{3}{2}\zeta(3) - 1 = 0.8030853\dots$ (known); $r = \frac{1}{2}$ gives $12(\zeta(3) - 1) - 1 = 1.4246828\dots$; and two equal touching balls give $\frac{21}{16}\zeta(3) - 1 = 0.5776997\dots$

Derivation. The image of an axial dipole in a rigid sphere is an axial dipole at the inverse point with strength scaled by $-(R/f)^3$ [23, 24]. Inverting about the contact point maps both spheres to parallel planes, so the image positions form an arithmetic progression. The strength ratios then telescope: $A_k = A_0 b^3/[a + k(b - a)]^3$. The added mass, $-(\rho/U) \oint \phi n_x dS$ over the ball, is a sum of Hurwitz-zeta values.

Verification. We reconstructed the flow from the images and checked both boundary conditions at 2000 random surface points. Brute-force surface integration, with no zeta functions used, gives, for example, 1.424682623 ($N = 6000$ images) against 1.424682838 for $a = 1, b = 2$. The truncation gap shrinks by a factor of about 4 per doubling of N . This result is probably a routine extension of classical image methods.

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